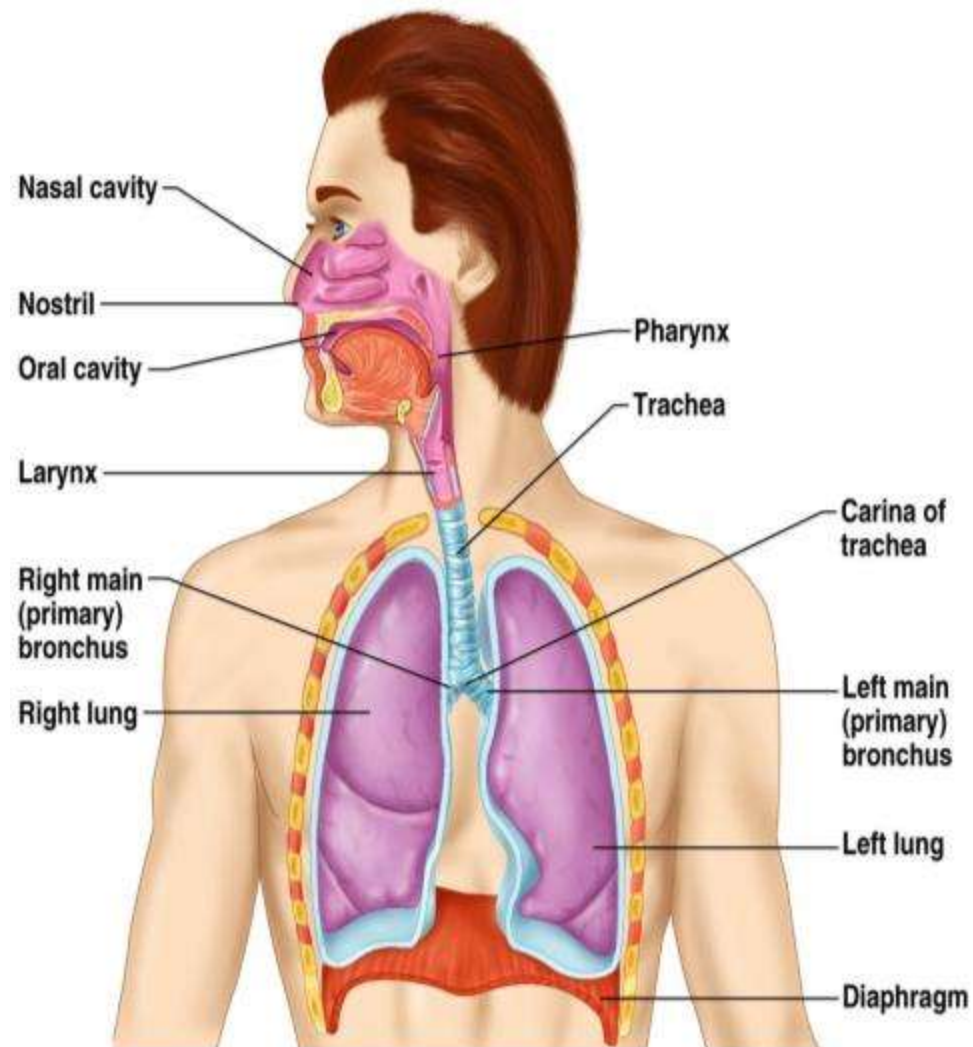


Respiratory system
By
Dr. Nali Abdulkader

Respiratory System

- Consists of upper and lower parts
- **The upper respiratory** include nose, nasal cavity, laryngopharynx and larynx
- **The lower respiratory** include, the trachea, bronchi, bronchial tree, and the lungs



The Nose

- The nose consists of the external nose and the nasal cavity, both of which are divided by a septum into right and left halves.

- **External Nose**

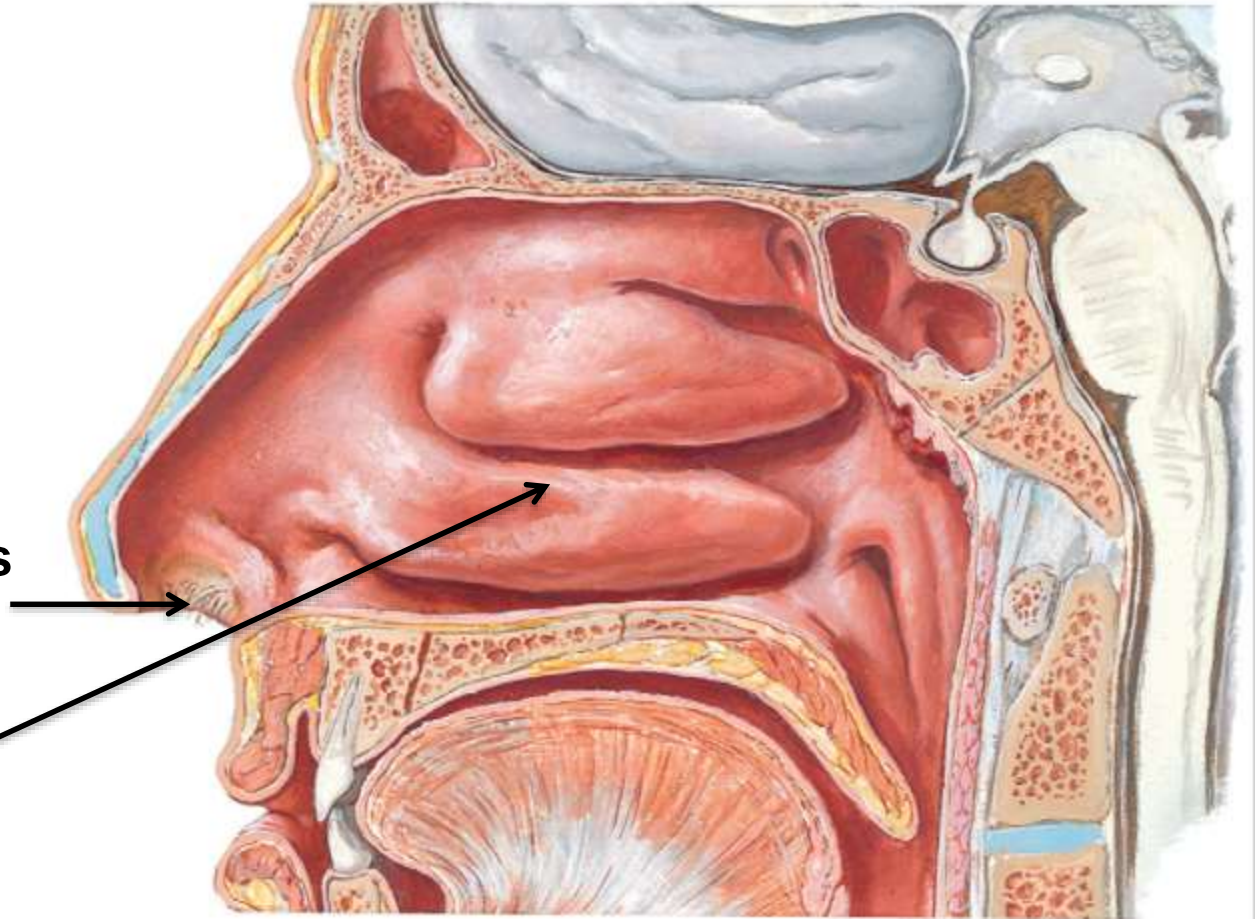
The external nose has two elliptical orifices called the **nostrils**, The lateral margin, the **ala nasi**, is rounded and mobile.

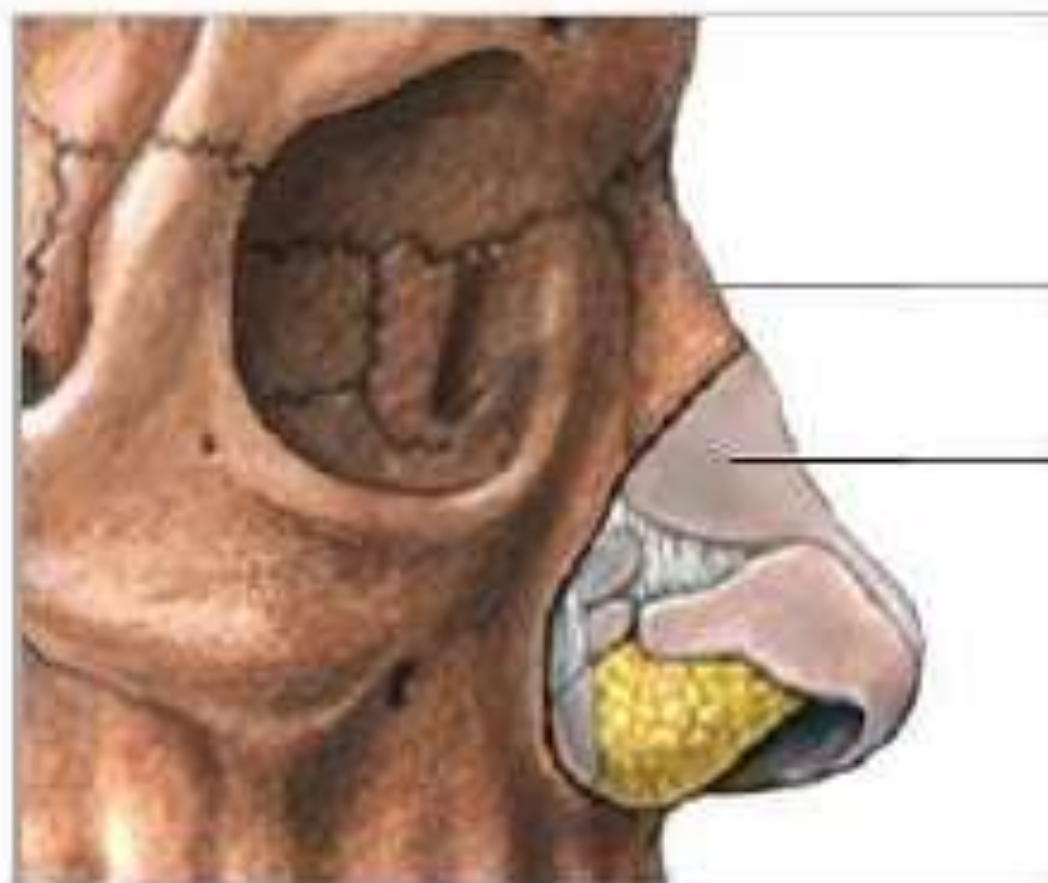
- **Nasal Cavity** The nasal cavity extends from the nostrils in **front**, to the posterior nasal apertures or choanae **behind**, where the nose opens into the nasopharynx.

Nose

External nasal openings
(nostrils or nares)

Internal nasal
(choanae).

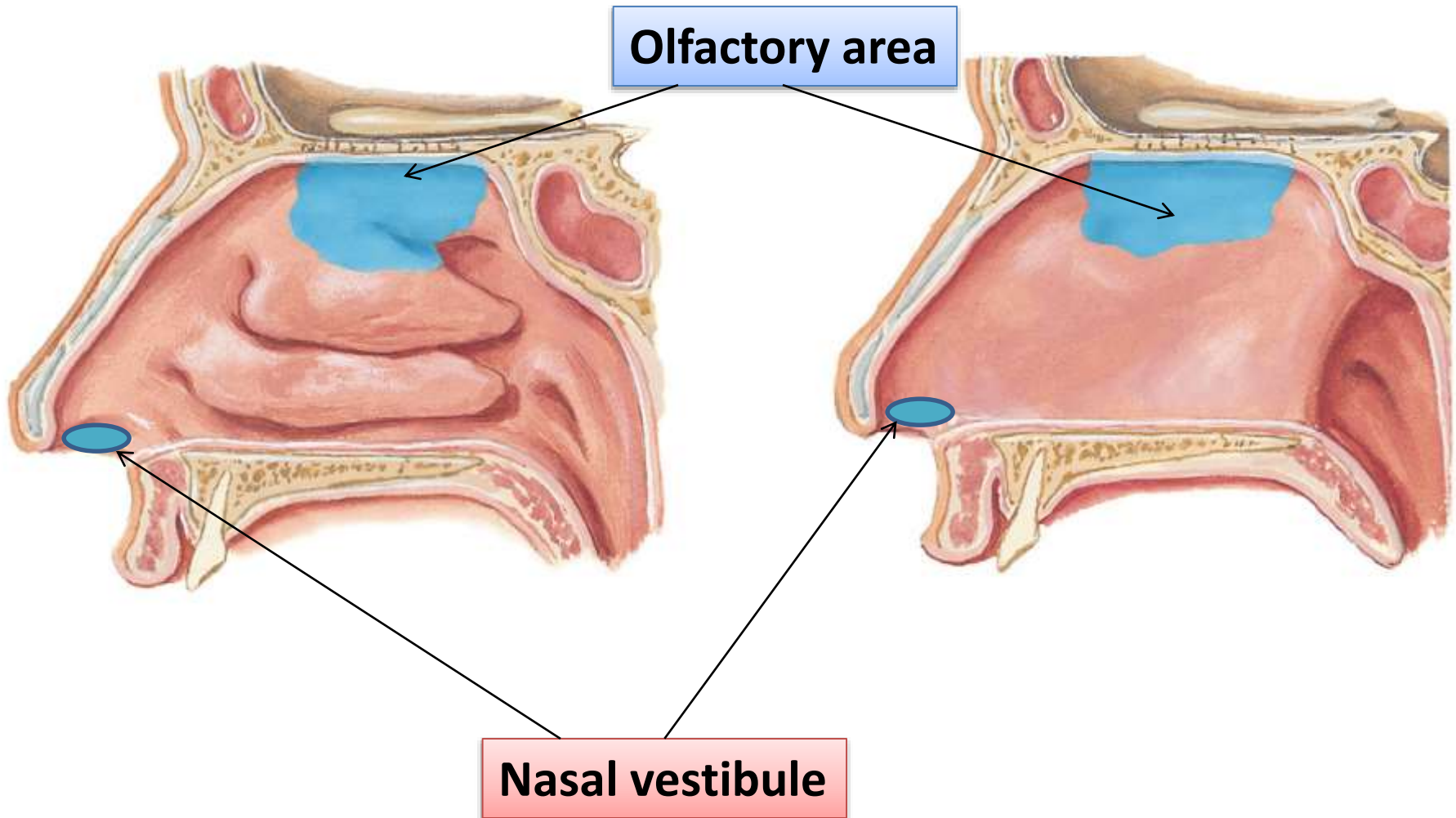




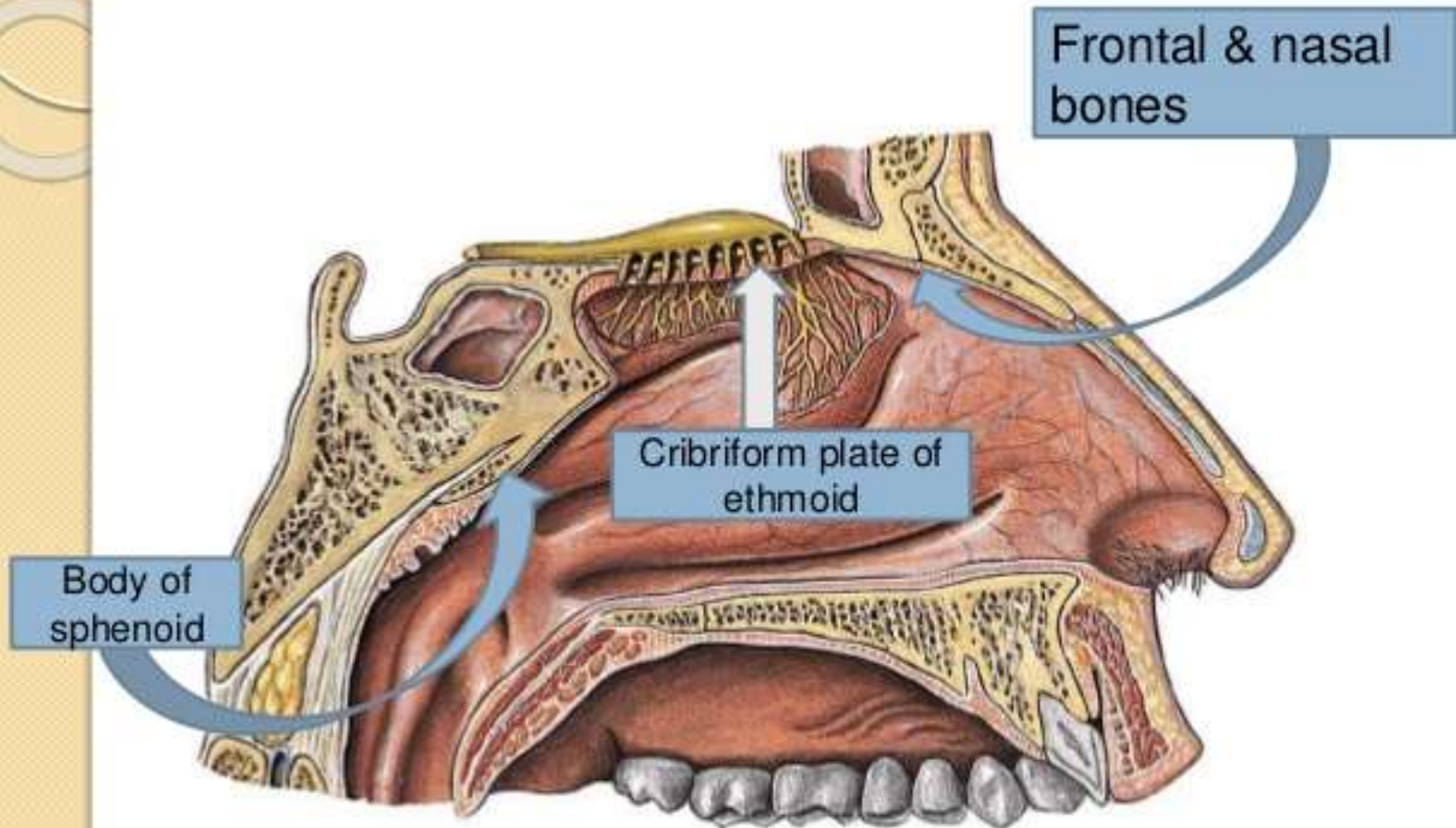
Bone

Cartilage

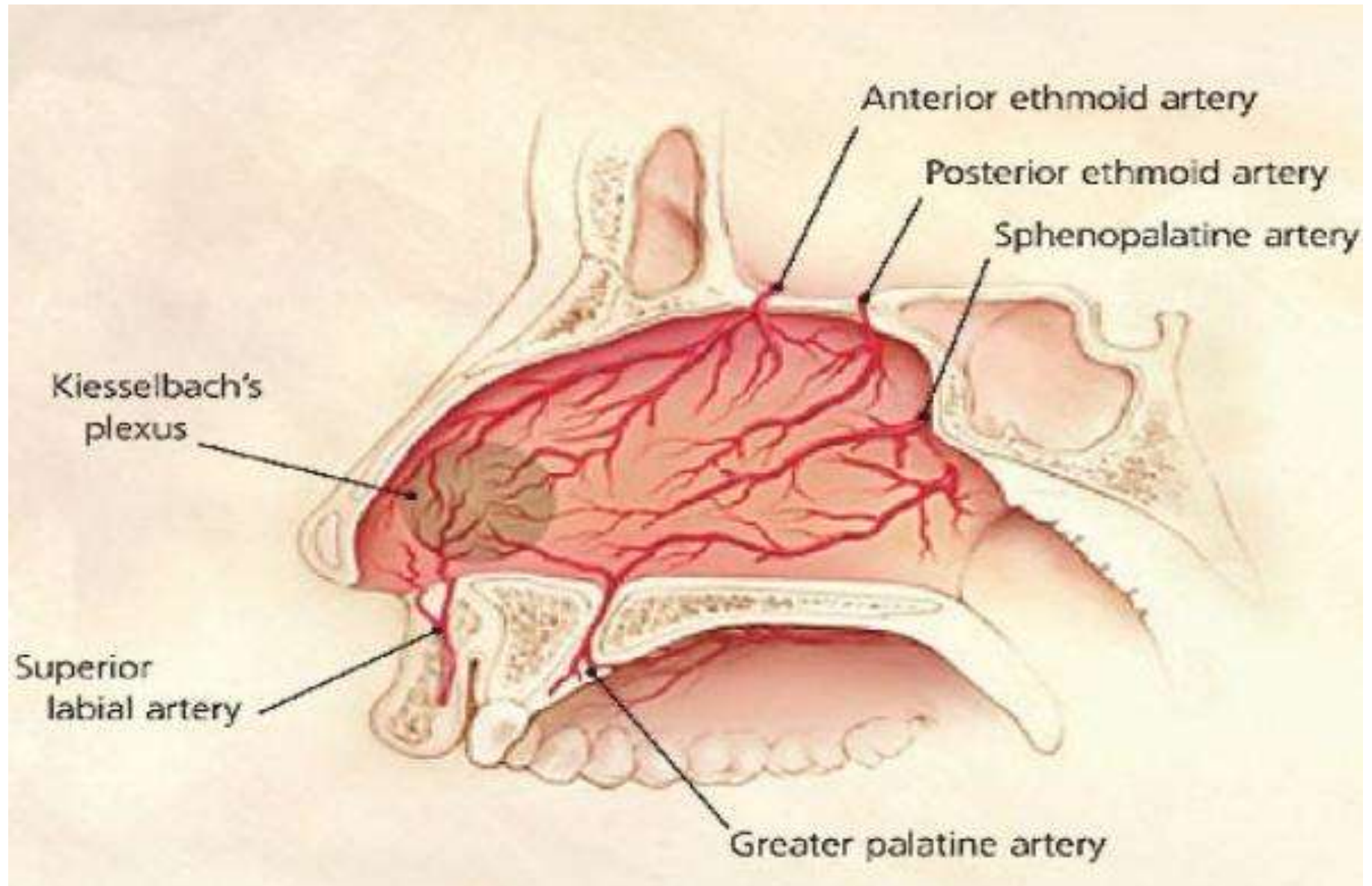
Regions of the nasal cavity



ROOF OF NASAL CAVITY



Little's area





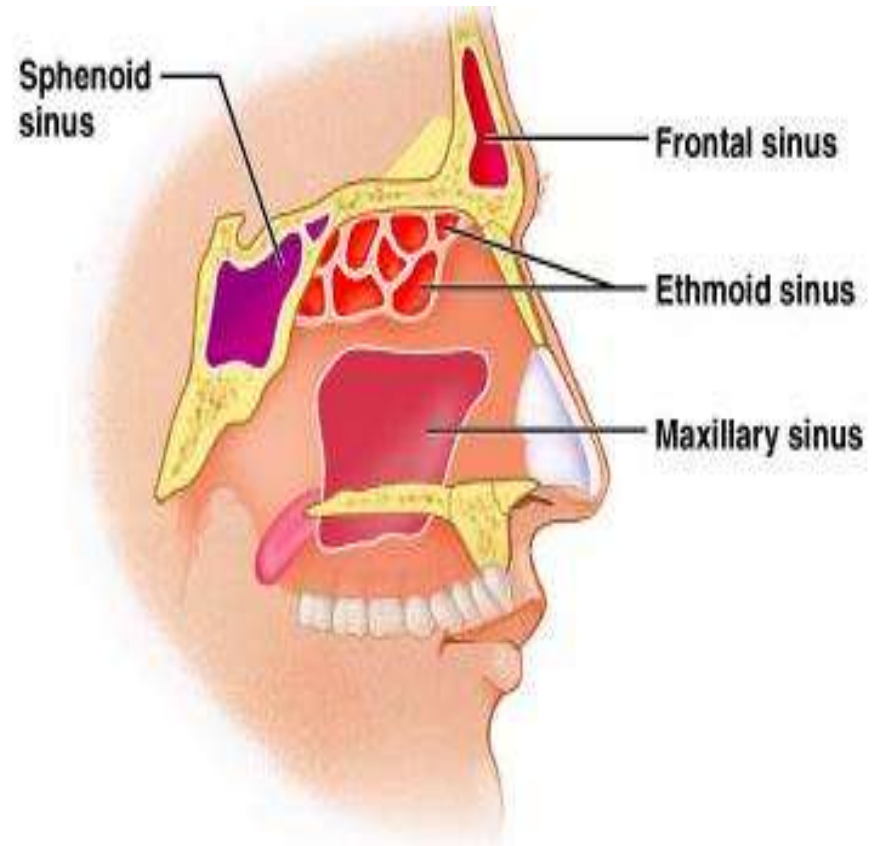
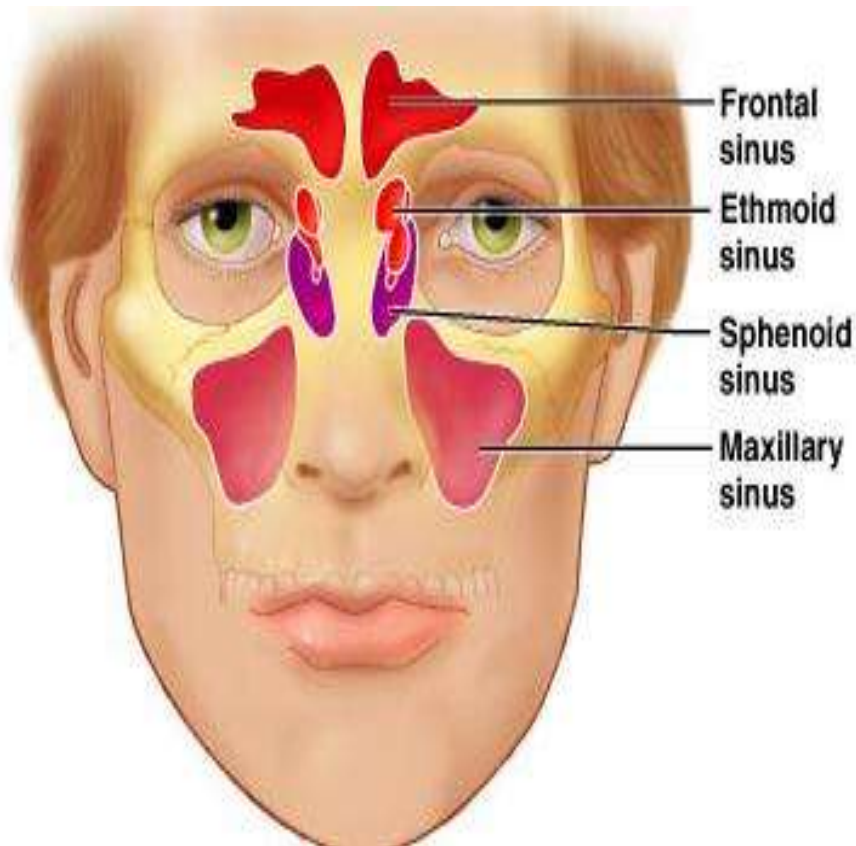
- Epistaxis
- Rhinorrhea
- Rhinitis

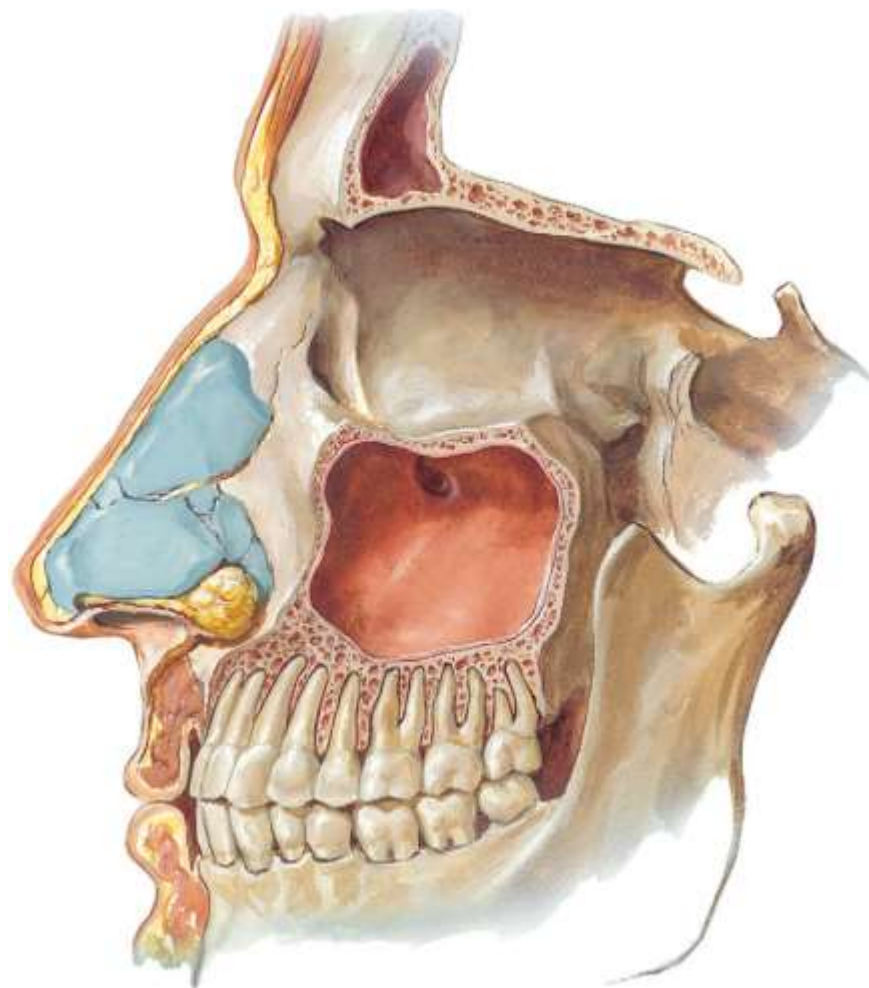
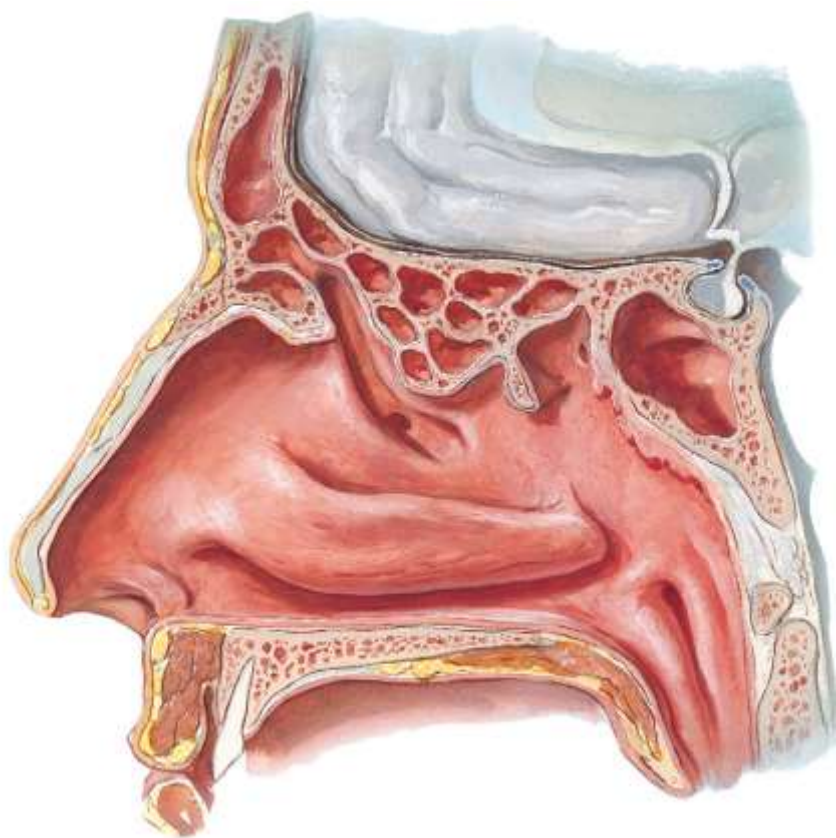




Paranasal sinuses

- Air cavities located in the bones of skull around the nose and open in the lateral wall of the nasal cavity.
- They include frontal, ethmoidal, sphenoidal and maxillary air sinuses.





The Pharynx

- 12-13 cm long muscular tube lies behind the oral, nasal and laryngeal cavities.
- Extends from the base of skull to the lower border of cricoids cartilage at the level of lower border of 6th cervical vertebra where it continues with esophagus.

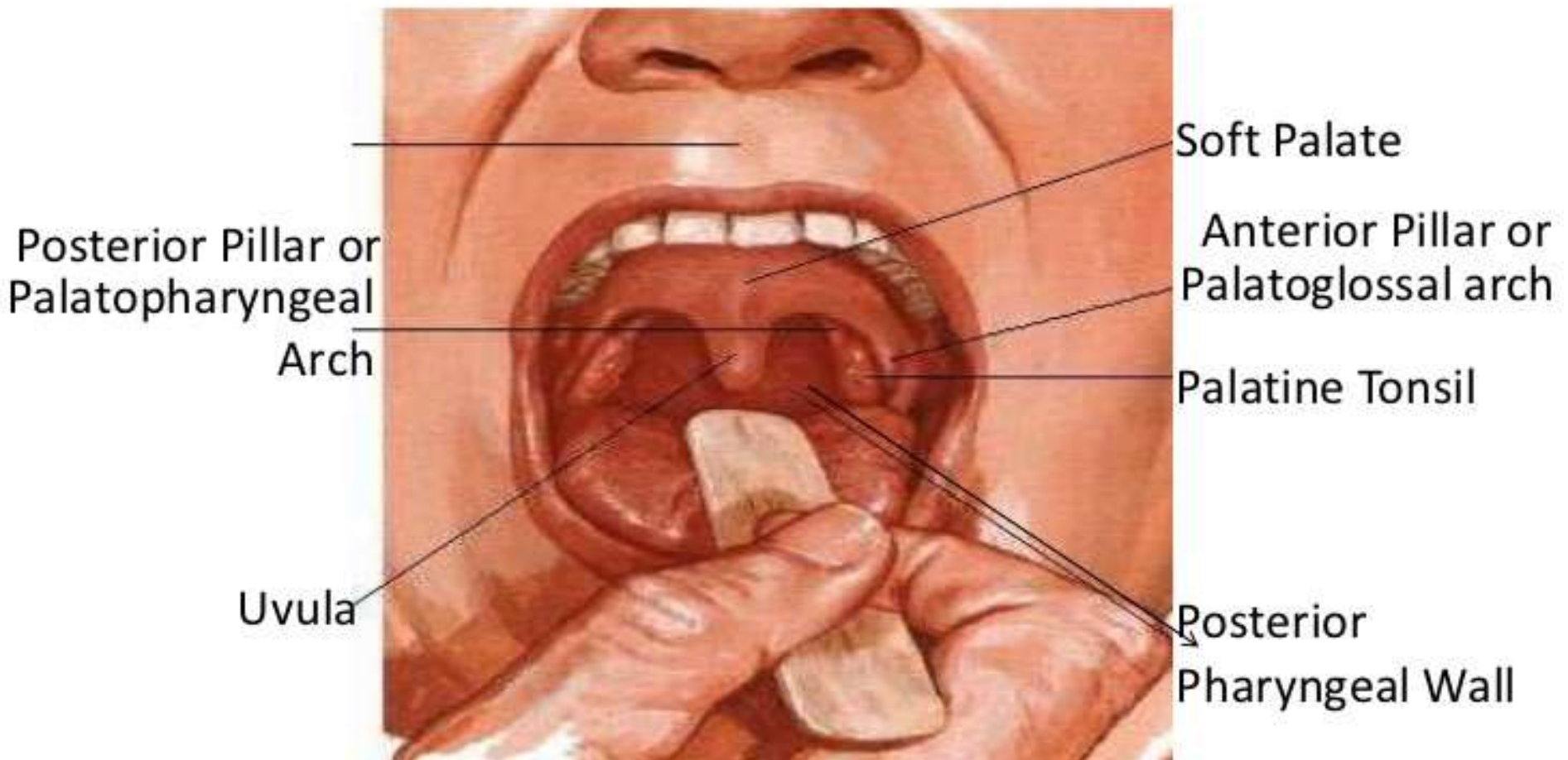
Parts of Pharynx

Naso-pharynx

- The upper part of pharynx lies behind the nasal cavity
- Each auditory tube opens in the lateral wall of nasopharynx

OROPHARYNX

It is the middle part of the pharynx situated behind the oral cavity.

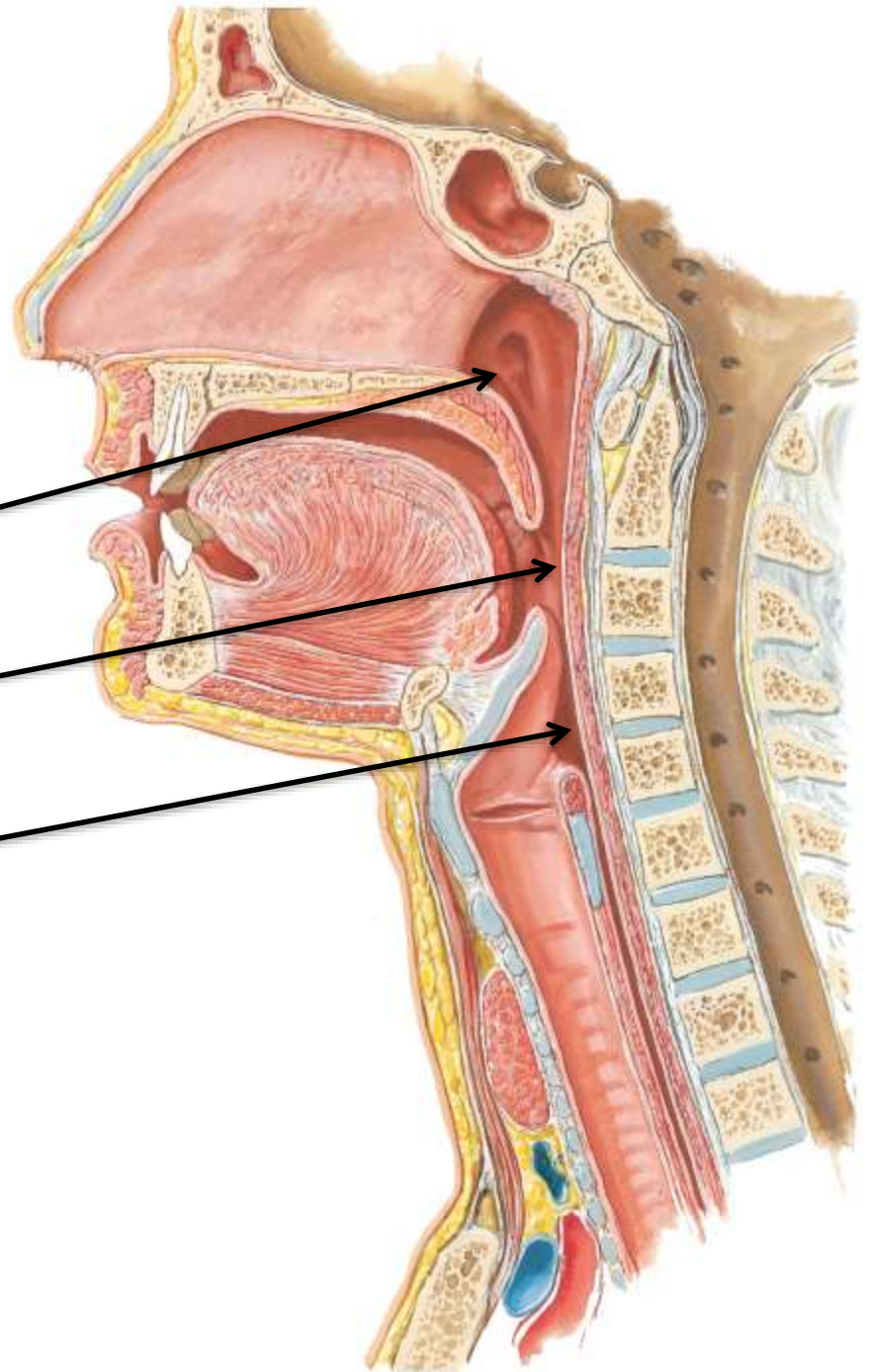


The Pharynx

- long muscular tube
lies behind the oral, nasal

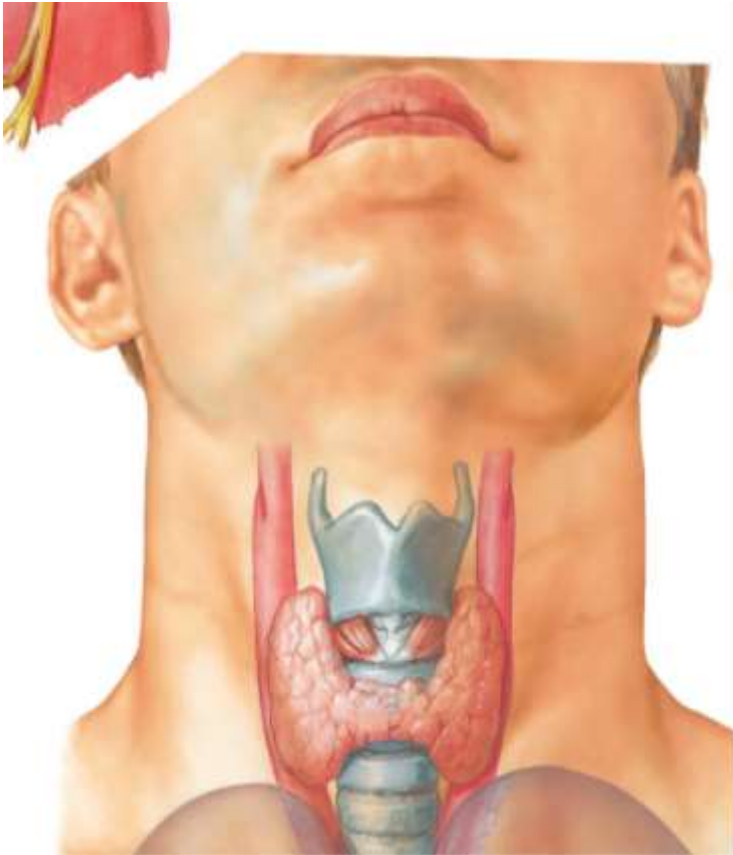
Parts of Pharynx

1. Naso-pharynx
2. Oro-pharynx
3. Laryngo-pharynx



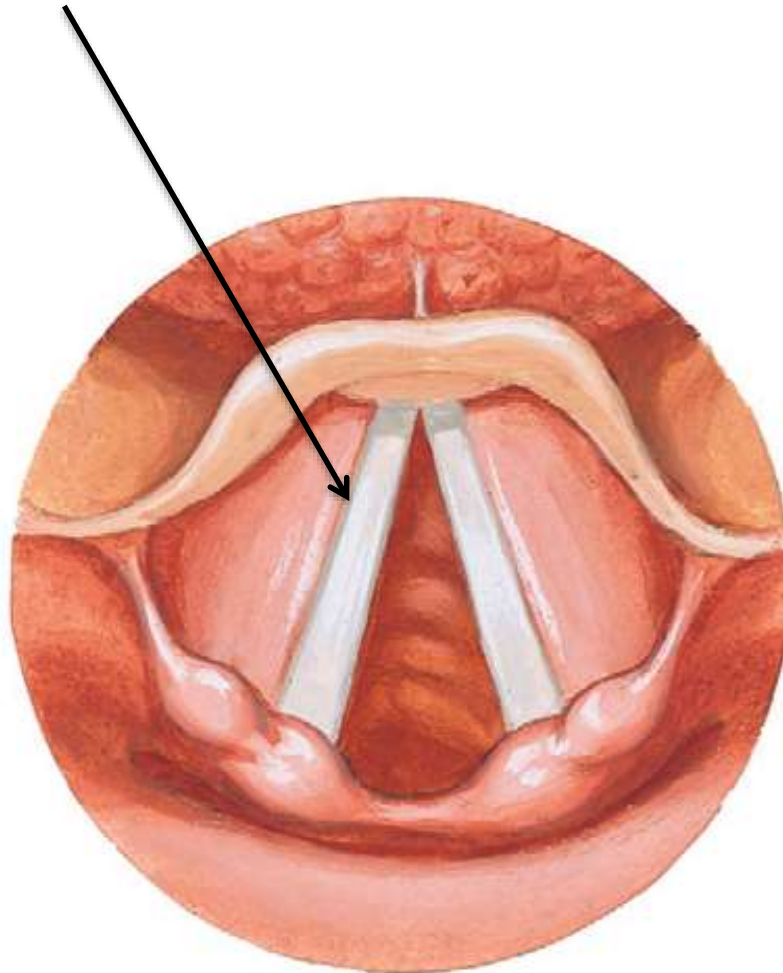
Larynx

- A phonation box and air passage, located in the middle anterior part of the neck.
- It is made of a number of cartilages connected together by membranes, ligaments, muscles



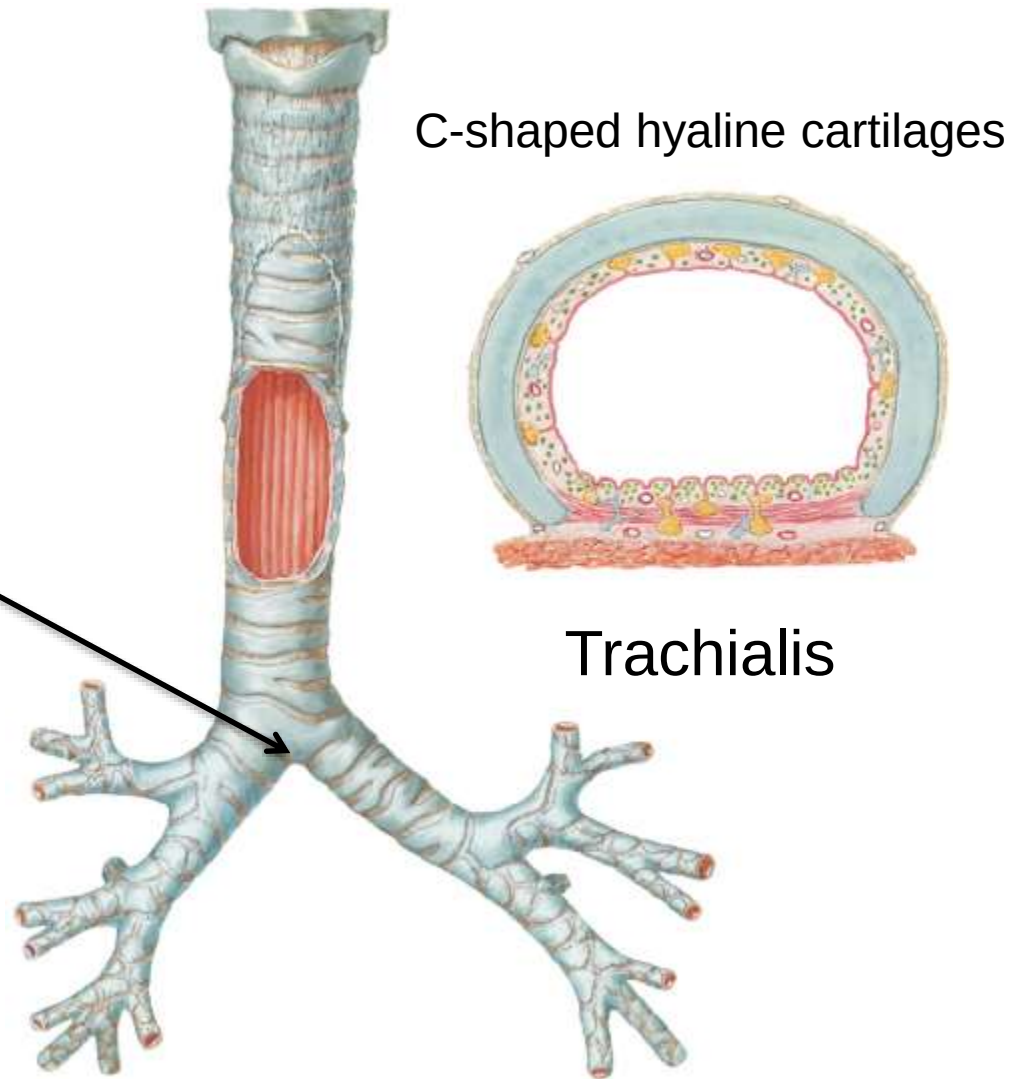
Cavity of Larynx

- Extends from inlet of larynx to the beginning of trachea
- In the central, middle part of laryngeal cavity, it contains two pairs of vocal folds or cords



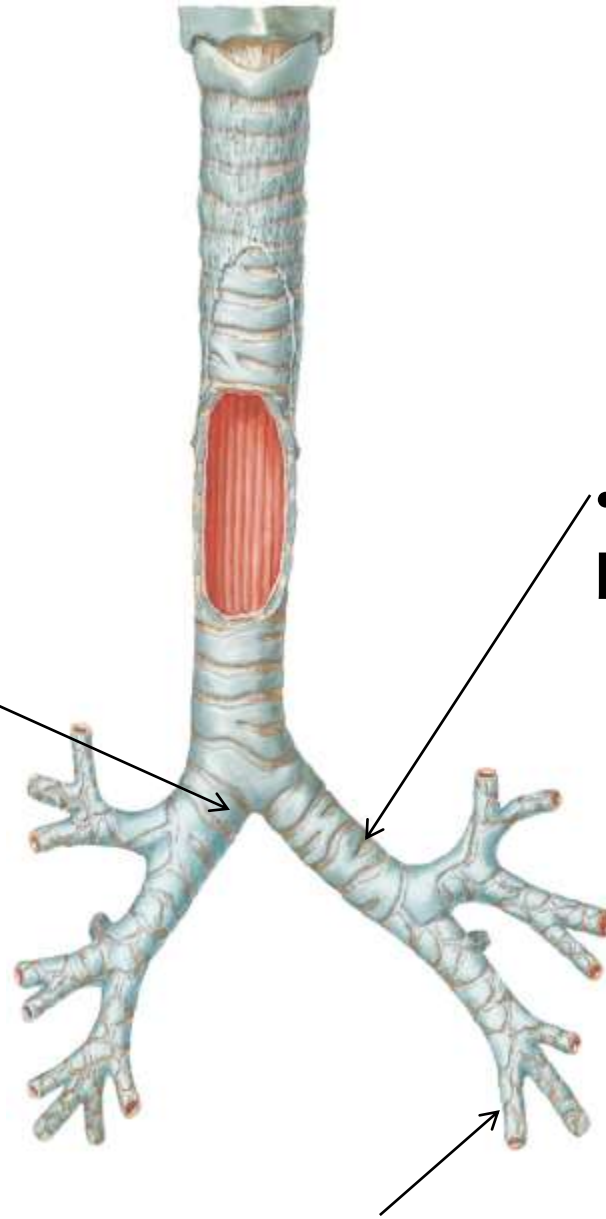
Trachea

- The trachea begins at the laryngeal outlet and terminates at the level of the sternal angle, by dividing into the right and left bronchi (**carina = bifurcation of trachea**)
- It is 10 cm long having a fibromuscular wall supported by a number of C-shaped hyaline cartilages.



Bronchi and Bronchial Tree

- The right and left principle bronchi begins at the level of sternal angle
- The **right principle**



•The **left principle bronchus**

**Tertiary bronchus
(Bronchopulmonary Segment)**

Lungs

- upper lobe
- middle lobe
- lower lobe

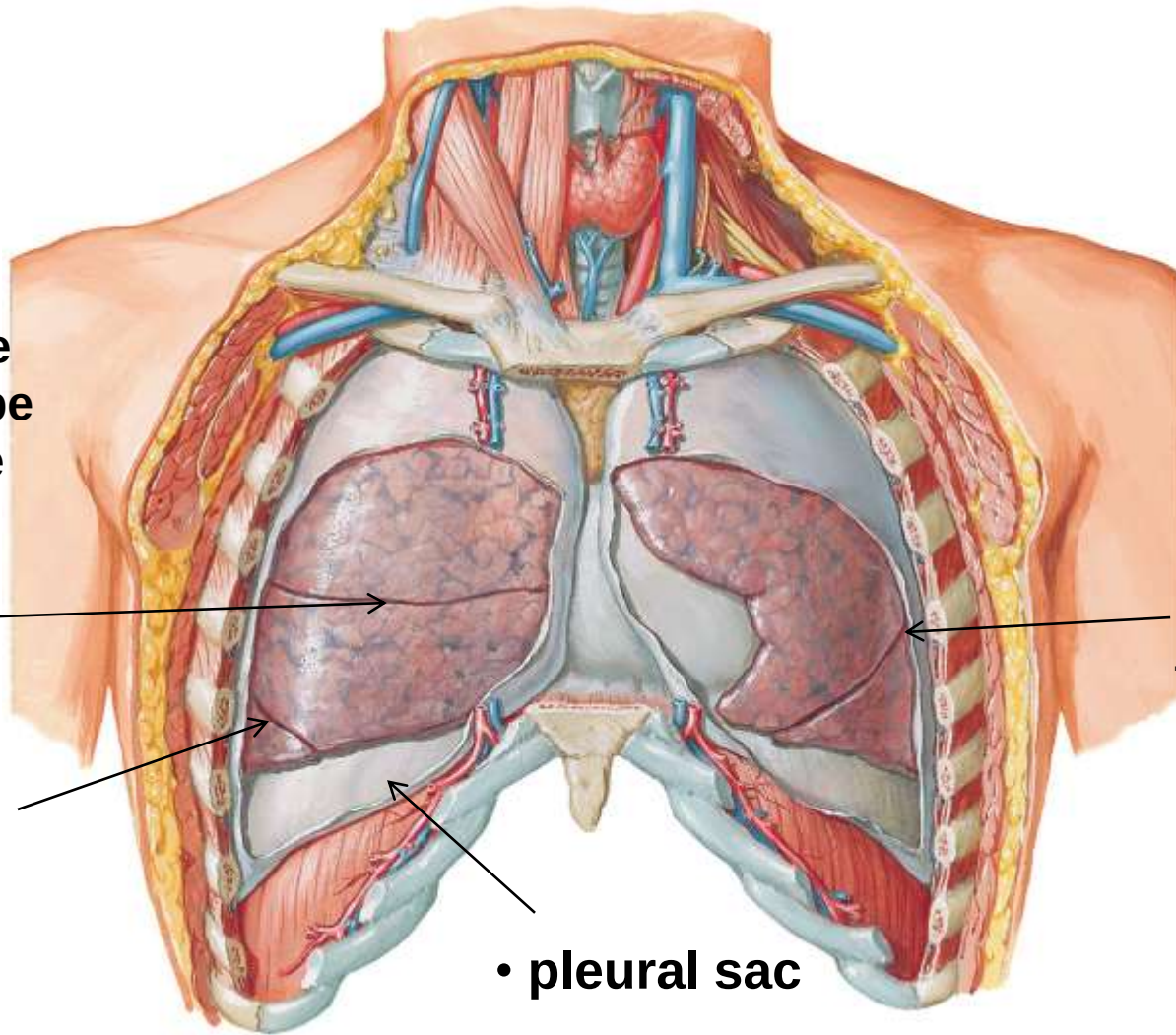
• transverse
fissures.

• Oblique
fissures.

• pleural sac

- upper lobe
- lower lobe

• Oblique
fissure.



Gross anatomy of lungs

- There are a pair of lungs in the thoracic cavity.
- Each lung is covered by pleura. Protected by thoracic skeleton.
- Each lung has an apex, a broad base or inferior surface, sharp anterior border, and sharp inferior border.
- Each lung has sterno-costal surface, diaphragmatic surface and mediastinal surface.
- Right lung is shorter than the left due to right lobe of liver pushing the right lung superiorly.
- There are 3 lobes in right lung and 2 lobes on left lung.
- There are two fissures in right lung, oblique and horizontal fissures.
- Left lung has only one oblique fissure.

Difference between Rt & Lt Lung

Right Lung	Left Lung
It has 2 fissures and 3 lobes	Only 1 fissure and 2 lobes
Anterior border is straighter	Anterior border is interrupted by the cardiac notch
Larger and heavier weighs about 700g	Smaller and lighter, weighs about 600g
Shorter and broader	Longer and narrower

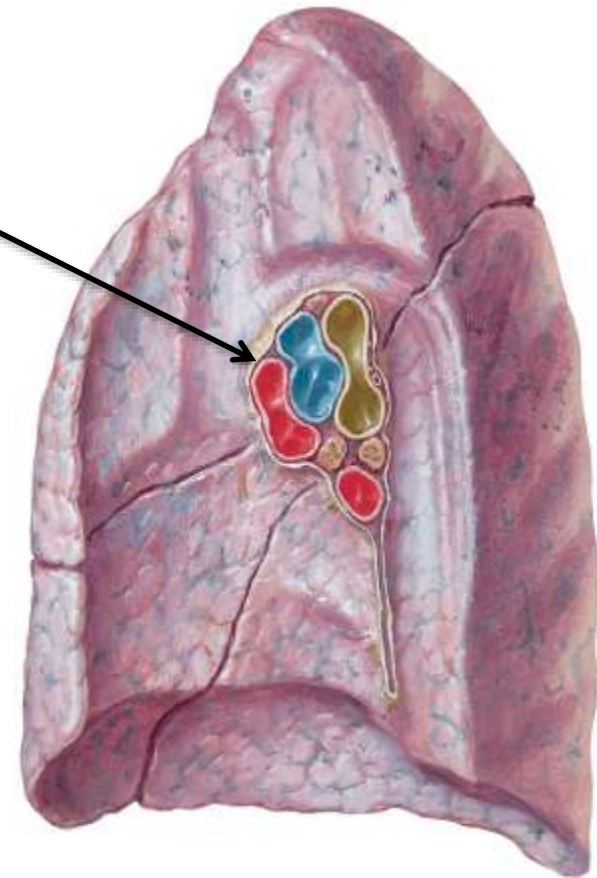
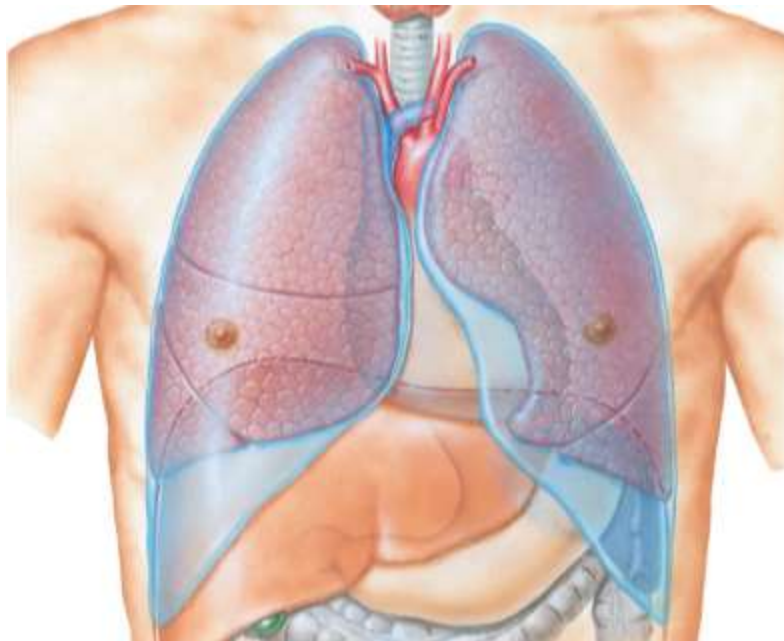
❖ **The costal surface** of the lung is large, smooth, and convex. It is related to the costal pleura, which separates it from the ribs, costal cartilages, and innermost intercostal muscles.

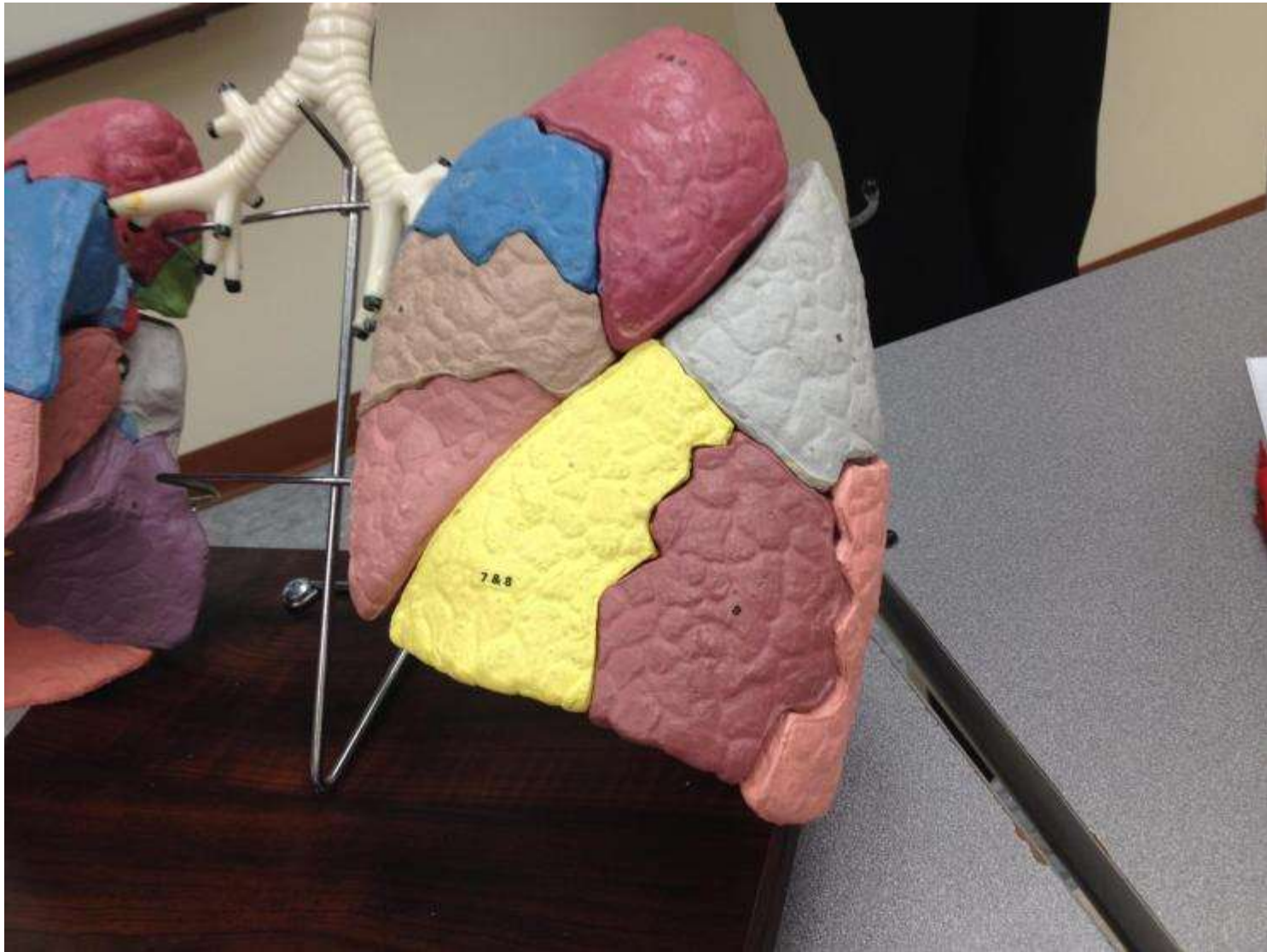
❖ **The mediastinal surface** of the lung is concave because it is related to the middle mediastinum, which contains the pericardium and heart. The mediastinal surface includes the hilum, which receives the root of the lung.

❖ **The diaphragmatic surface** of the lung, which is also concave, forms the base of the lung, which rests on the dome of the diaphragm

The hilum of the lung

is a wedge-shaped area on the mediastinal surface of each lung through which the structures forming the root of the lung pass to enter or exit the lung.



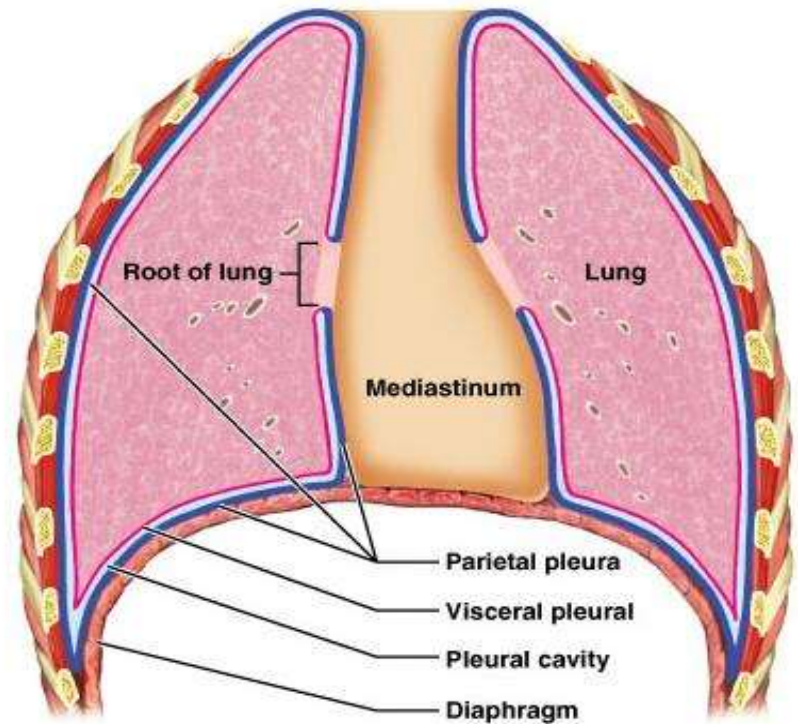


Pleura

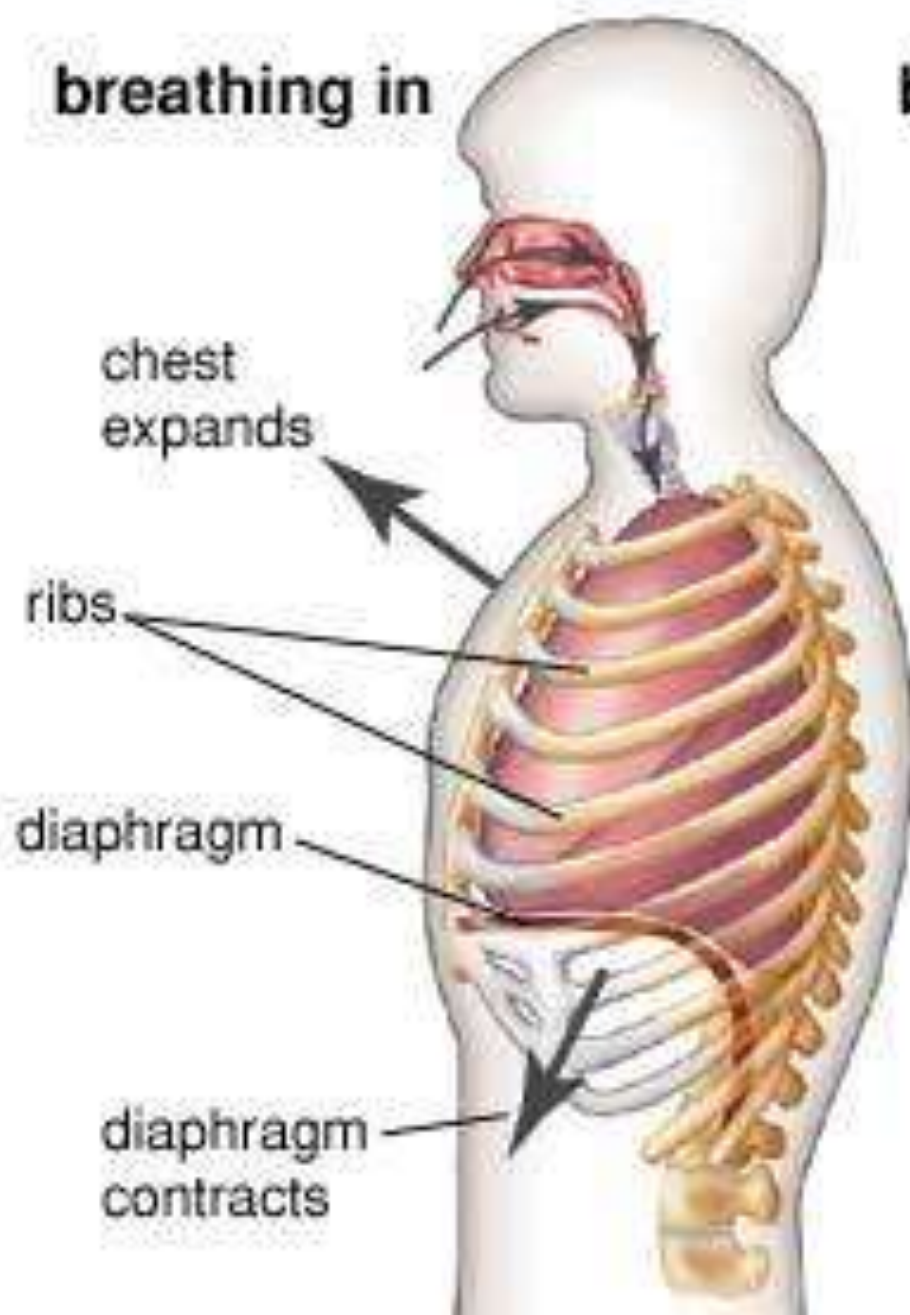
- Serous sac investing the lungs.
- Consists of outer **parietal** and inner **visceral** layers separated by a small space called **pleural cavity**.



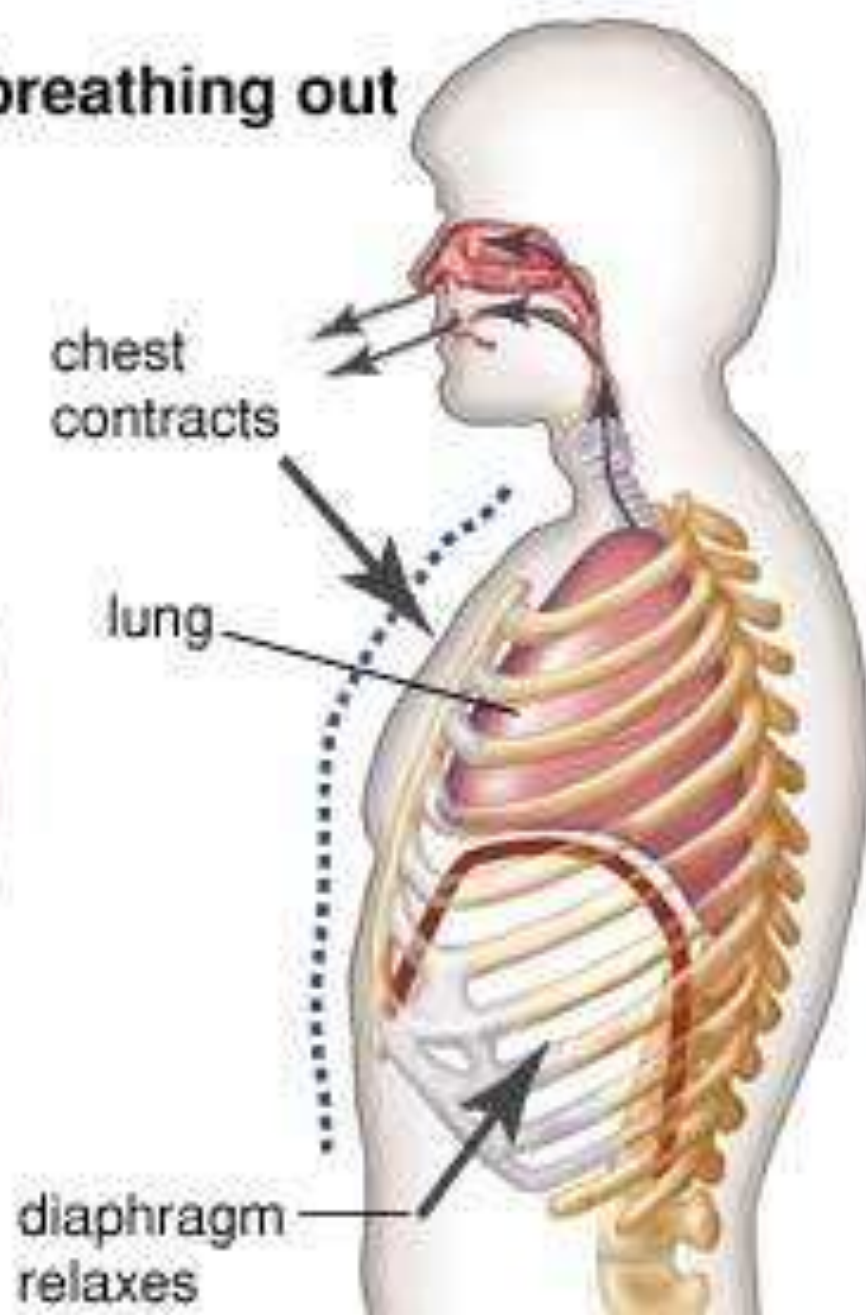
-Pleuritis
-Pneumothorax
-Haemothorax



breathing in

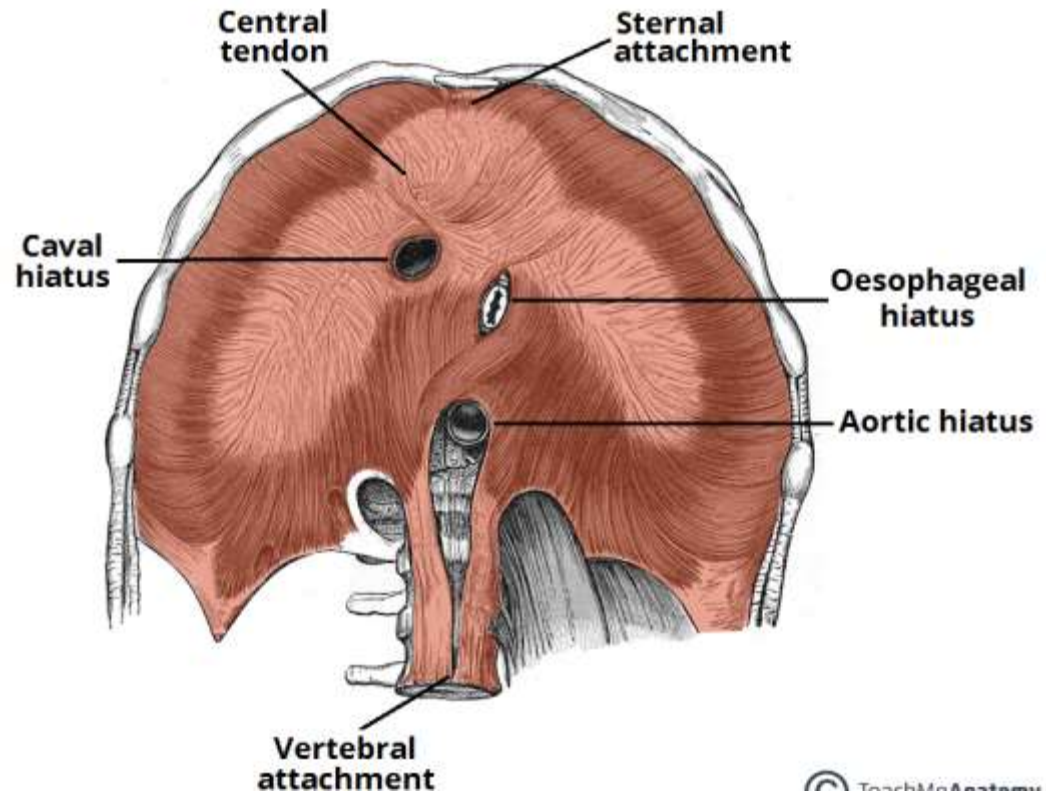


breathing out



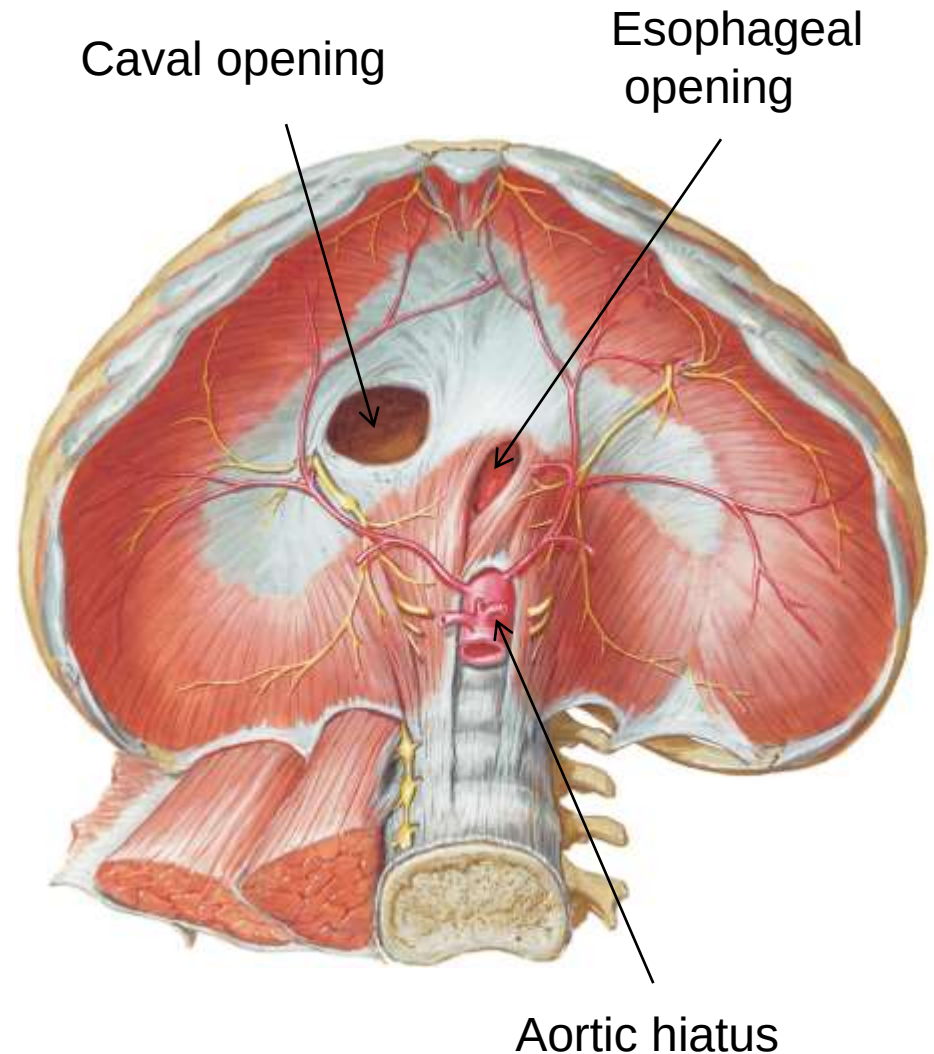
Diaphragm

- ❖ Is the musculotendinous it has two parts
 - *Muscular part fixed*
 - *Central part movable*
 - The fibers of the muscular part radiated in the center forming central tendon and it has no bony attachments



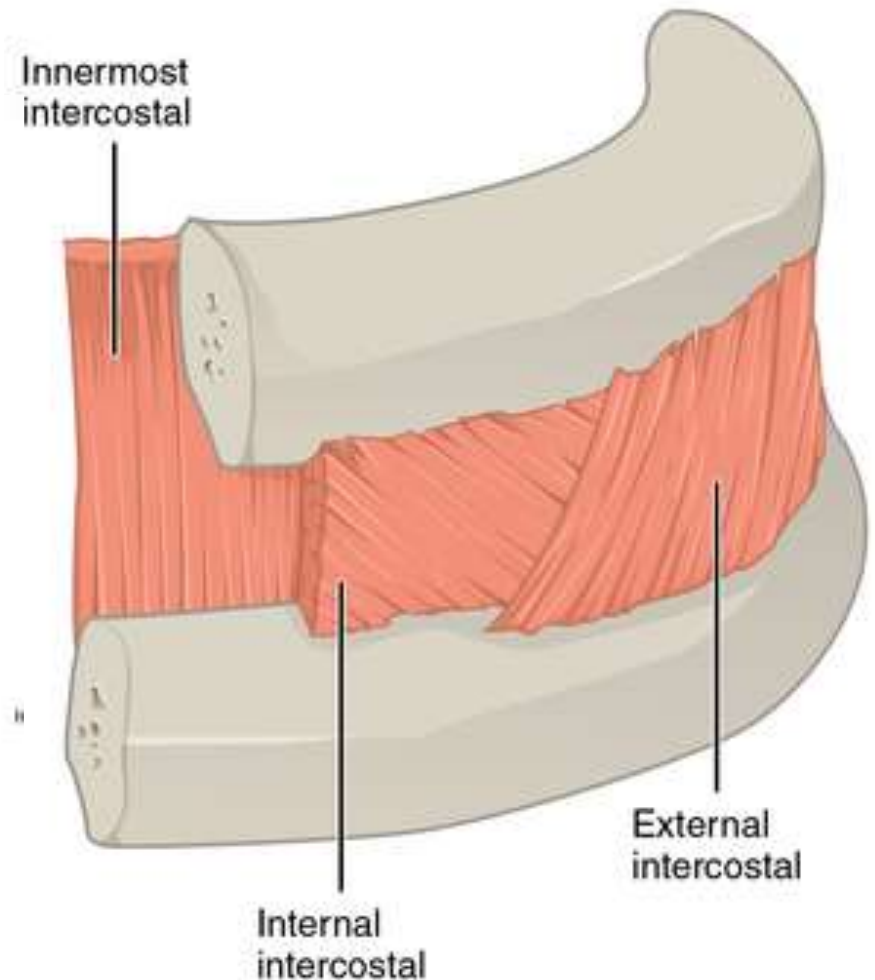
Opening of the diaphragm

- **Aortic hiatus** at the level of T12 for aorta
- **Esophagus hiatus** at the level of T10 for esophagus
- **Caval opening** for IVC at the level of T8



Intercostal muscles

- **External intercostal muscle**
- **Internal intercostal muscle**
- **Innermost intercostal muscle**
 - Intercostalis intima
 - Subcostalis
 - Sternocostalis { transversus thoracis }





Questions